Exponential Models

Name _____

Algebra 2 Lesson 6-2 (fast track) Date _____ Period _____

Learning goal: I can write an exponential model for money that earns compound interest or continuously compounded interest, and use it to find the value of the account.

Language goal: I can explain why more compounding periods give a larger account value, using the words principal, interest rate, compounding periods, compounded continuously, and natural base e.

On the quiz: (Savvas numbers) (1) A town's yearly decrease as a rate per quarter and (2) a point on $y = a \cdot b^x$ from two points: full Savvas quiz only. You will (3) build $A(t) = P\left(1 + \frac{r}{n}\right)^{nt}$ for an account compounded monthly (Example 2); (4) find the value of an account compounded continuously (Example 4); (5) bonus: compare simple, continuous and yearly interest on $8,000 and say which option is linear and which are exponential (Examples 2 and 4, Practice).

Example 2: Understand Compound Interest

Compound interest is interest earned on interest already deposited. **Compound interest formula:** $A = P\left(1 + \frac{r}{n}\right)^{nt}$, where P = the initial **principal** invested, r = annual **interest rate** as a decimal, n = number of **compounding periods** per year, and A = the value of the account after t years.

Tamira invests $5,000 in an account that pays 4% annual interest. **A.** How much will there be in the account after 3 years if the interest is compounded annually, quarterly, or monthly? ($P = 5000$, $r = 0.04$, $t = 3$.)

	n	**Compound interest formula**	**After 3 years**
Annually	1	$A = 5000\left(1 + \frac{0.04}{1}\right)^{3(1)} = 5000(1.04)^3$	$5,624.32
Quarterly	4	$A = 5000\left(1 + \frac{0.04}{4}\right)^{3(4)} = 5000(1.01)^{12}$	$5,634.13
Monthly	12	$A = 5000\left(1 + \frac{0.04}{12}\right)^{3(12)} \approx 5000(1.003333)^{36}$	$5,636.36

As the number of compounding periods increases, the amount in the account also increases: the more often interest is added, the earlier it earns interest of its own.

B. What is the annual growth rate for each possible account? Use the Power of a Power rule, $(b^n)^t = b^{nt}$, to rewrite each model with exponent t. Growth factor minus 1 is the rate.

	Rewrite with exponent t	**Annual growth rate**
Annually	$A = 5000\left(1 + \frac{0.04}{1}\right)^{1t} = 5000(1.04)^t$	4%
Quarterly	$A = 5000\left(1 + \frac{0.04}{4}\right)^{4t} = 5000\left((1.01)^4\right)^t \approx 5000(1.0406)^t$	4.06%
Monthly	$A = 5000\left(1 + \frac{0.04}{12}\right)^{12t} \approx 5000\left((1.003333)^{12}\right)^t \approx 5000(1.0407)^t$	4.07%

Quarterly: $1.0406 - 1 = 0.0406 = 4.06\%$. This yearly rate is the **effective annual interest rate**.

Say it: Compounded monthly, the account has _____ compounding periods per year instead of 1, so its value after 3 years is _____ than with annual compounding, because ...

Try It 2: Compound interest

$3,000 is invested in an account that earns 3% annual interest, compounded monthly.

$P =$ _____ $r =$ _____ $n =$ _____ $A =$ _____ $\left(1 + \frac{\quad}{\quad}\right)^{\quad t}$

a. What is the value of the account after 10 years? 100 years?

After 10 years:

After 100 years:

b. What is the effective annual interest rate of the account?

Hint: rewrite the model with exponent t, as in Part B.

Example 4: Find Continuously Compounded Interest

The natural base e. Put $1 in at 100% interest for one year, compounded n times: the value is $\left(1 + \frac{1}{n}\right)^n$. For $n = 1$ it is 2; $n = 100$, 2.704813829; $n = 100{,}000$, 2.718268237. As n keeps growing, the value approaches the irrational number $e = 2.718281828459\ldots$, the **natural base**. Compounding that never stops is **compounded continuously**.

Continuously compounded interest formula: $A = Pe^{rt}$, where P = the principal, e = the natural base, r = annual interest rate as a decimal, and A = the value after t years.

Regina invests $12,600 in an account that is compounded continuously, at a 3.2% interest rate for a 12-year term. What is the value of the account after 12 years? Round your answer to the nearest dollar.

1. $P = 12{,}600,\ \ r = 0.032,\ \ t = 12.$

2. $A = Pe^{rt} = 12{,}600\, e^{0.032(12)}$

3. Multiply the exponent first: $0.032(12) = 0.384$, so $A = 12{,}600\, e^{0.384}$.

4. Use the e^x key: $e^{0.384} \approx 1.468145$, so $A \approx 18{,}498.63$.

To the nearest dollar, the value is $18,499.

Calculator trap. When you evaluate $e^{0.032(12)}$, either simplify 0.032(12) as 0.384 first, or type parentheses, `e^(0.032*12)`, so that e is raised to the entire product, not just the first factor.

Say it: The account is _____, so I use $A =$ _____. The base e is the _____, about _____.

Try It 4: Continuously compounded interest

You invest \$125,000 in an account that earns 4.75% annual interest, compounded continuously.

$P =$ _____ $r =$ _____ $A =$ _____ $e^{\text{_____}t}$

a. What is the value of the account after 15 years?
b. What is the value of the account after 30 years?

a. After 15 years:

b. After 30 years:

Why did you use $A = Pe^{rt}$ and not $A = P\left(1 + \frac{r}{n}\right)^{nt}$? Use the words *compounded continuously* and *natural base*.

Practice: Three ways to save

Lena has \$5,000 from a summer job to save for 8 years. She compares three accounts.

Option A: 4% simple interest
Option B: 3.5% interest compounded continuously
Option C: 3.75% interest compounded yearly

Simple interest is paid on the principal only: $A = P(1 + rt)$. It adds the same amount, Pr, every year.

a. Write a model for each option and fill in the table. Round to the cent.

	Model	**After 1 year**	**After 8 years**
A (simple)	$A =$ _____	_____	_____
B (continuous)	$A =$ _____	_____	_____
C (yearly)	$A =$ _____	_____	_____

b. After 8 years, to the nearest dollar, Option B will be worth \$_____ more than Option A and \$_____ less than Option C.

c. Option A is a _____ function, while Options B and C are _____ functions. Which option gained the most in the first year? Which gains faster over time? Use your table.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
LWE	Evaluate exponential functions (warm-up)	Do Now: calculator powers and the e key
YJW ★	Compound interest: word problems	Quiz 3 and 5 (Example 2)
5GU ★	Continuously compounded interest: word problems	Quiz 4 and 5 (Example 4)
KLF ★	Describe linear and exponential growth and decay	Quiz 5 (bonus), last parts: linear or exponential

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Who will have more?

Adam and Jacinta open an account on the same day. Each starts with \$8,000 for 20 years. Adam's interest is compounded quarterly at 1.25%. Jacinta's interest is compounded continuously at 1.25%. Who do you predict will have more money in their account after 20 years? Explain your reasoning.

Adam: $A =$ _____ Jacinta: $A =$ _____

_____ will have more, because interest that is _____ is added more often than interest with only _____ compounding periods per year, so ...

Before the quiz, finish these sentences.

The _____ is the money you put in at the start; r is the annual _____, written as a decimal.

More _____ per year means a _____ account value.

When interest is _____, I use $A = Pe^{rt}$, where $e \approx$ _____ is the _____.